

Ali Mamaghani | Curriculum Vitae

✉ amamaghani@ucsd.edu • in ali-mamaghani • 🔑 aAP3asAAAAAJ

Education

University of California San Diego

Ph.D. in Electrical and Computer Engineering

GPA-3.9/4

Advisor: Prof. Dinesh Bharadia

La Jolla, CA, USA

2024-2029 (expected)

Sharif University of Technology

Masters in Electrical Engineering (Communication Systems)

GPA-19.11/20.00(4/4)

Thesis: Applications of LLMs in a Cloud-Native 5G Platform

Advisors: Prof. Babak Hossein Khalaj and Prof. Navid Nikaein (EURECOM)

Tehran, Iran

2022-2024

Sharif University of Technology

Bachelors in Electrical Engineering (Communication Systems)

GPA-19.15/20.00(3.99/4)

Tehran, Iran

2018-2022

Allameh Helli High School (NODET)

Mathematics and Physics Diploma.

NODET: National Organization for Development of Exceptional Talents

Tehran, Iran

2014-2018

Young Scholars Club

IPhO Team Selection Camp

Tehran, Iran

Jun 2017- Sep 2017

Research Interests

- Communication Networks
- Machine Learning
- LLM Agents
- Large Language Models
- Reinforcement Learning
- ML for Networking

Research Experience & Publications

Question Answering over Telecom Log Systems

Jan 2025 - Present

Research Project

- Supervisor: Dinesh Bharadia
- Proposed PRISM, a two-stage framework combining causal-semantic log pruning and LLM-based reasoning for interpretable 5G system analysis.
- Designed a log-pruning pipeline that uses a RAG approach to condense millions of log lines into query-aligned templates, delivering up to 30% higher factual accuracy and ~66% lower compute time.
- Developed a hypothesis-verification reasoning algorithm integrated with a 3GPP-derived knowledge graph, enabling log causal reasoning, diagnosing issues: buffer overflow, RACH failures, bad channels, HARQ retransmission, etc.
- Validated the framework on over-the-air srsRAN testbeds and created a multi-scenario 5G log dataset covering both SDR and commercial RU deployments for reproducible evaluation.
- Publications:
 - A. Mamaghani, Q. Zheng, U. Ghosh, V. Rajagopalan, S. Shakkottai, et al., "Log Pruning for Efficient Q&A in 5G Networks," *Proc. 2nd ACM Workshop on Open and AI RAN*, pp. 36-42, 2025.
 - O. Kamatani, S. Saruwatari, S. Seshasayee, A. Mamaghani, D. Bharadia, "LLM-5GMAC: Performance Optimization in O-RAN Split 7.2 Using LLM-Based MAC-Layer Log Analysis," *Proc. 2nd ACM Workshop on Open and AI RAN*, pp. 1-7, 2025.
 - Q. Zheng, U. Ghosh, A. Mamaghani, S. Shakkottai, D. Bharadia, "PHOENIX: Protocol State Machine Recovery from Events in 5G Logs," *Proc. 16th ACM Workshop on Wireless of the Students, by the Students, and for the Students (S3)*, 2025.

Benchmarking LLM Software-Engineering Agents on Telecom Codebases

2026

Research Project

- Collaborators: Prof. Vijay K. Shah (NC State), Prof. Dinesh Bharadia
- Built TeleSWEBench, the first commit-driven benchmark measuring how well automated software-engineering agents can navigate and modify mathematically rigorous wireless stacks such as srsRAN 5G.
- Mined real developer commits from the srsRAN repository and distilled them into 734 structured test cases with executable unit tests across three difficulty tiers (Easy, Medium, Difficult).
- Proposed TeleJudge, a hierarchical LLM-as-a-Judge framework scoring agent outputs at file level and aggregating verdicts holistically via context and semantic similarity, complementing unit-test evaluation.

- Evaluated AIDER, OpenHands, and Claude Code driven by Qwen3, GPT-OSS, Gemma 4, Kimi, and Qwen-Coder 2.5, revealing weak fault localization and functional correctness (best tools ship only ~25% of changes).
- *Publication:*
 - P. Gajjar, A. Mamaghani, D. Bharadia, V. K. Shah, "TeleSWEBench: A Commit-Driven Benchmark for Evaluating LLM-Powered Software Engineering in Telecommunications," *arXiv:2606.05001*, 2026.

Tiny-Twin: CPU-Native Full-Stack Digital Twin for NextG

Jan 2025 - Present

Research Project

- *Supervisor:* Dinesh Bharadia
- Built a CPU-native, full-stack digital twin of NextG cellular networks enabling reproducible, hardware-free experimentation and large-scale evaluation of RAN algorithms.
- *Publications:*
 - A. Mamaghani, U. Ghosh, S. Shakkottai, D. Bharadia, I. K. Jain, "Tiny-Twin: A CPU-Native Full-stack Digital Twin for NextG Cellular Networks," *IEEE Int. Symp. on Spectrum Innovation (DySPAN)*, pp. 1–8, 2026.
 - A. Mamaghani, U. Ghosh, S. Shakkottai, D. Bharadia, I. K. Jain, "Demo: Tiny-Twin – A CPU-Native Full-Stack Digital Twin for NextG Cellular Networks," *IEEE Int. Symp. on Spectrum Innovation (DySPAN)*, 2026.

Edgeric for OAI

Sep 2024 - Jan 2025

Research Project

- *Supervisor:* Dinesh Bharadia
- In this project I integrate Edgeric (A real-time radio intelligence controller) into the OAI 5G RAN, trying to increase user throughput using RL algorithms, and providing real-time monitoring for the fine-grain measurements of UE. This leads to application-aware optimization of RAN.

Applications of LLMs in a Cloud-Native 5G platform from Eurecom

Apr 2023 - Aug 2024

Master's Thesis Project

- *Supervisors:* Prof. Hossein Khalaj and Prof. Nikaein (from Eurecom)
- In this project, first, I build a 5G network using Kubernetes and 5G components from OAI, then I provide AI power for different aspects of the network including providing explanations for the output of tests and errors, correcting configuration files using GPT3.5-turbo in a feedback loop and fine-tuning GPT3.5-turbo to generate configurations and develop a chatbot as an assistant.
- *Publications:*
 - A. Mamaghani, A. Nourian, N. Mohtaram, A. Shokrani, S. M. Nasiri, et al., "LLM for 5G: Network Management," *IEEE Int. Conf. on Machine Learning for Communication and Networking (ICMLCN)*, 2024.
 - A. Majlesara, A. Majlesi, A. Mamaghani, A. Shokrani, B. H. Khalaj, "5G Network Automation Using Local Large Language Models and Retrieval-Augmented Generation," *arXiv:2511.21084*, 2025.

Implementation of 4G and 5G mobile Networks with centralized controller

Aug 2021 - Jun 2022

Bachelor's Thesis Project

- *Supervisor:* Prof. Hossein Khalaj
- In this project, I managed to implement a 4G network and different scenarios of 5G Networks(SA, NSA). Furthermore, I used O-RAN RIC and slicing xApp in this network. In these implementations, I used OAI, FlexRAN/FlexRIC from Mosaic5G, and Open5GS platforms.

Teaching Experience

CSE 123: Computer Networks, Instructor: Prof. Schulman

Spring 2026

- Held discussion and Q&A sessions, and graded Exams

Game Theory, Instructor: Prof. Ashtiani

Spring 2024

- Designed problem sets, held Q&A sessions, and graded homework

Numerical Optimization, Instructor: Prof. Babaie-Zadeh

Fall 2023

- Designed problem sets, held Q&A sessions, and graded homework

Probability and statistics, Instructor: Prof. Ashtiani

Fall 2023

- Designed problem sets, held Q&A sessions, and graded homework of +20 students

Data Networks, Instructor: Prof. Pakravan

Fall 2022

- Designed problem sets, held Q&A sessions, and graded homework of +30 students

Random Processes, Instructor: Prof. Behnia

Spring 2022

- Held Q&A sessions, and graded homework of +20 students

Data Networks, Instructor: Prof. Pakravan

Spring 2022

- Designed problem sets, held Q&A sessions, and graded homework of +30 students

- Discrete-time Signal Processing**, Instructor: Prof. Shamsollahi Fall 2021
 ○ Designed problem sets and held Q&A sessions with +30 students
- Circuits Theory**, Instructor: Prof. Shamsollahi Fall 2020
 ○ Designed problem sets and held Q&A sessions with +30 students
- Electromagnetics**, Instructor: Prof. Memarian Spring 2020
 ○ Designed course website, designed problem sets, held Q&A sessions, and graded homework of +120 students

Honours and Awards

- **NSF Student Travel Grant**, IEEE DySPAN 2026
- **NSF POSE Travel Grant**, OPen6G+AI School 2026
- **Graduate Fellowship winner**, UC San Diego 2024
- **Ranked 2nd** among Electrical Engineering-communication systems master's Students Nov 2023
- **Ranked 5th** among 176 Electrical Engineering Students 2019 -2022
- Runner-up for Best BSc Project Award. Apr 2022
- **Exceptional Talents Award**, Sharif University of Technology Feb 2022
 Granted to the top 3% of students; confers direct admission to the MSc program, waiving the national entrance exam.
- **Silver Medal** in the 31st National Physics Olympiad 2017
 Ranked among the top 40 nationally in a multi-stage competition open to all high school students in Iran; qualified for the IPhO Team Selection Camp.
- **Silver Medal**, International Mathematics Competition (IMC), Iran National Round 2013
 Awarded to top performers in a nationwide mathematics competition open to all middle school students in Iran.

Presentations and Invited Talks

- **OpenAirInterface workshop**: Tiny-twin: A CPU-Native Full-stack Digital Twin for 5G 2025
 Hosted by UT Austin, openairinterface
- **MobiCom OpenAI workshop**: Log Pruning for Efficient Q&A in 5G Networks 2025
 The 31st Annual International Conference on Mobile Computing and Networking Nov. 4-8, 2025, Hong Kong, China
- **6G Summit Abu Dhabi**: 5G Network Automation Using LLM Models and RAG 2024
 Hosted by IIT, UAE
- **Ericsson AI/ML/6G Weekly**: LLM for 5G: Network Management 2024
 Hosted by Amin Azari, Sweden
- **ICMLCN**: LLM for 5G: Network Management 2024
 IEEE International Conference on Machine Learning for Communication and Networking, Sweden

Selected Projects

Game Theory

DA matching Algorithm Implementation Spring 2023

- Assigning patients with special disease types to hospitals using Python. Each hospital has a disease preference and each disease has a hospital preference.

Numerical Optimization

Implementation of Optimization Algorithms Fall 2022

- Implementing line search based on strong Wolfe condition, steepest descent, Newton, and BFGS algorithms in MATLAB.

Massive Data Analysis

City Traffic Analysis Fall 2022

- Finding hot spots and rush hour of a city based on camera data of cars in a city.

Deep Learning

Sentiment Detection Fall 2022

- Fine-tuning multi-modal models and transformers to detect sentiment based on text and image.

Queuing Theory

Simulation of M/M/m queue Fall 2022

- Simulating different Queuing scenarios including queues that contain positive and negative signals using Python.

Cloud Software Mobile Network

Routing Algorithm Implementation

Spring 2022

- Implementing dynamic routing with the Dijkstra algorithm on an SDN controller(ODL) in Python.

Advanced Data Networks

MIPS Processor Implementation

Spring 2022

- Implementing Multi-Cycle and Single Cycle architecture of MIPS Processor using Verilog.

Information Theory

Distributed Coding

Fall 2021

- Presenting and providing a term paper on distributed coding.

Data Networks

Simplified RPC Protocol

Fall 2021

- Implementing a simplified RPC protocol using socket programming and multi-threading in Python.

Machine Learning

Biometric Authentication

Spring 2021

- Biometric Authentication using ECG signals. Preparing noisy signals with MATLAB filtering toolbox and using SVM for classification.

Convex Optimization

Convex Optimization in Economics

Spring 2021

- Convex optimization on an economic problem using MATLAB and CVX.

Computer and Microprocessor architecture

MIPS Processor Implementation

Spring 2020

- Implemented Multi-Cycle and Single Cycle architecture of MIPS Processor using Verilog.

Programming Skills

MATLAB & Simulink

- Proficient in MATLAB programming (signal processing, CVX, digital filter design, and implementing machine learning algorithms and neural networks)
- Proficient in using Simulink to simulate communication systems and control systems.

Python Programming

- Proficient in programming using Python and experienced in working with libraries (CvxPy, sklearn, SciPy, TensorFlow, PyTorch, Spark, Pandas) and implementing learning algorithms using Python.

Other programming and tools: C/C++, go, bash, Kubernetes, ODL, NS3, GNS3, Git, Verilog, Assembly Mininet, Linux, Docker, Cuda

Selected Coursework

University of California San Diego

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|--------------------------------------|---|
| ○ Information Theory (A+) | ○ Computer Security(A+) |
| ○ Modern Communication Networks(A+) | ○ Networked Services(A+) |
| ○ Principles of Wireless Networks(A) | ○ Parallel Computation(A) |
| ○ Linear Algebra and Application(A) | ○ Machine Learning for Physical Applications(A) |

Sharif University of Technology

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|--|---|
| ○ Queuing Theory(20/20 - Graduate) | ○ Digital Communication(20/20) |
| ○ Advanced Data Networks(19.5/20 - Graduate) | ○ Convex Optimization(19.5/20) |
| ○ Game Theory(20/20 - Graduate) | ○ Communication Systems(19.6/20) |
| ○ Information Theory(19.1/20 - Graduate) | ○ Digital Signal Processing(20/20) |
| ○ Numerical Optimization(19.8/20 - Graduate) | ○ Linear Control Systems(20/20) |
| ○ Data Networks(19.5/20 - Graduate) | ○ Introduction to Machine Learning(20/20) |
| ○ Random Processes(20/20- Graduate) | ○ Probability and Statistics(19.7/20) |

Professional Service

- **Reviewer**, IEEE Transaction on Computers 2026
- **TPC Member and Reviewer**, ACM S3 Workshop (MobiCom) 2026
Workshop on Wireless of the Students, by the Students, and for the Students, co-located with ACM MobiCom

- **Reviewer**, IEEE Transactions on Dependable and Secure Computing 2026
- **Reviewer**, IEEE Future Networks World Forum 2026
- **Reviewer**, JMIR Formative Research 2026
- **Student Volunteer**, IEEE DySPAN 2026
- **Student Volunteer**, ACL 2026
Annual Meeting of the Association for Computational Linguistics

Mentorship

- **University of California San Diego**

Qingyuan Zheng, Wooil Jung, Nikhil Kuniyil — mentored on 5G log analysis and protocol state machine recovery; co-authored resulting publications.

- **Sharif University of Technology**

Alireza Shokrani, Seyed Mohsen Nasiri, Sepehr Kazemi Ranjbar, Parsa Hatami, Ahmadsreza Majlesara — mentored on LLM-based 5G network automation; co-authored resulting publications.

Language Skills

Persian: Native proficiency

English: Professional proficiency

Spanish: Elementary Proficiency